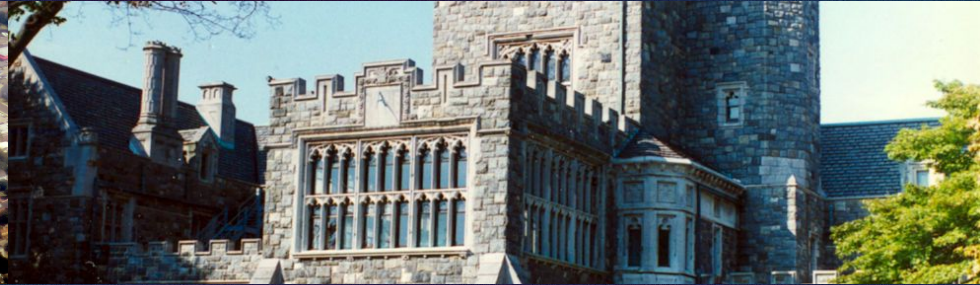


Is Suburban Road Density Linked With Poverty?



Contents

1. Background
2. Methodology
3. Data analysis
4. Results
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6. Conclusions



Background

Much of America lives in suburban communities outside of major inner cities - and thus it is useful for us to assess what features lead to prospering suburbs as opposed to failing ones.

This study sought to determine if road density was correlated with poorer neighborhoods in suburbs. With the needs of suburbs being different than those of other communities, it was hypothesized that rather than bringing goods and services, dense roads would lead to more undesirable suburbs.



Methodology

Location of Study - Nassau County of New York State.

Nassau County was chosen due to several unique features. It is a county with significant size, including a large population of ~ 1.4 million.

Though it is one of the wealthiest counties in the United States, there is large discrepancies in wealth between communities, particularly between the richer north coast and poorer south coast. This division between communities would make it easier to discern if a correlation involving wealth was present.

Lastly, Nassau County is notable for being nearly entirely suburban. It was speculated that the correlation hypothesized in suburban communities may not exist in rural areas, so a county that nearly exclusively contained suburbs was ideal.



Methodology

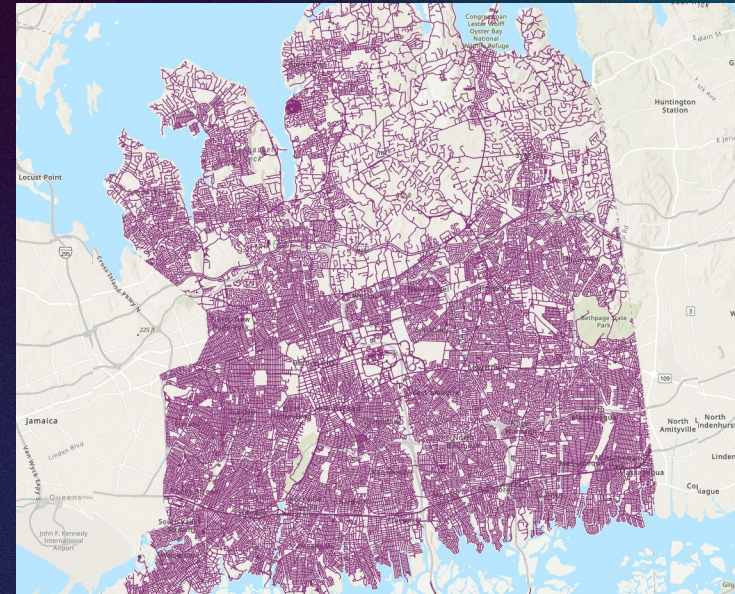
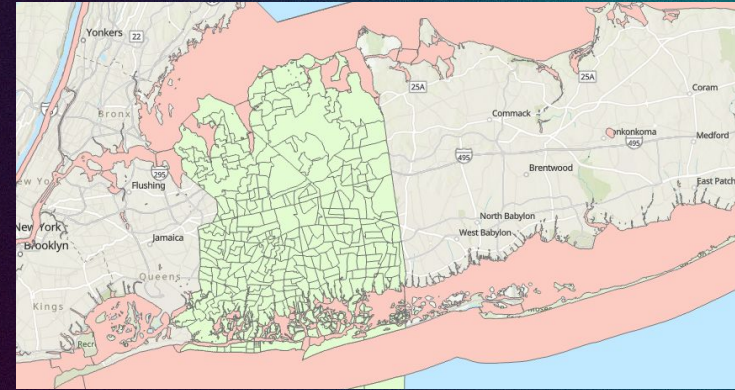
Datasets -

- Nassau County Census Tracts and tract-based table of Household Income in Nassau County, from U.S. Census
- Water Bodies of New York, from New York State GIS
- Roads of New York, from New York State GIS

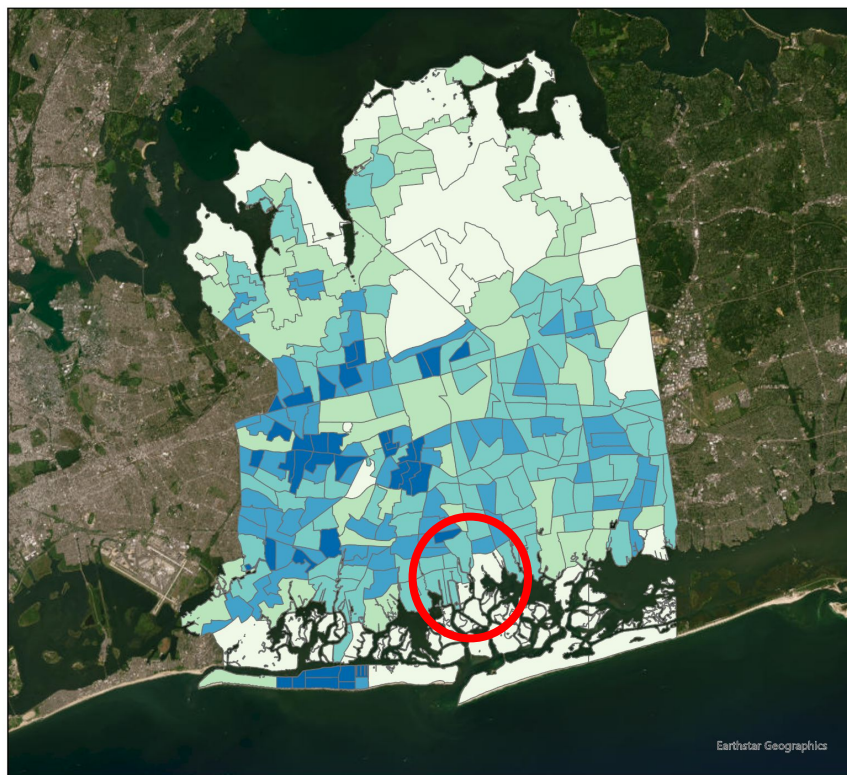
Processing -

Tracts were isolated to purely their land area by subtracting water. Household income was then joined to tracts.

Roads were isolated to purely residential roads (speed limits under 55 mph) within Nassau County. Roads were then added to tracts as well, their length summed to create a total road length per county.

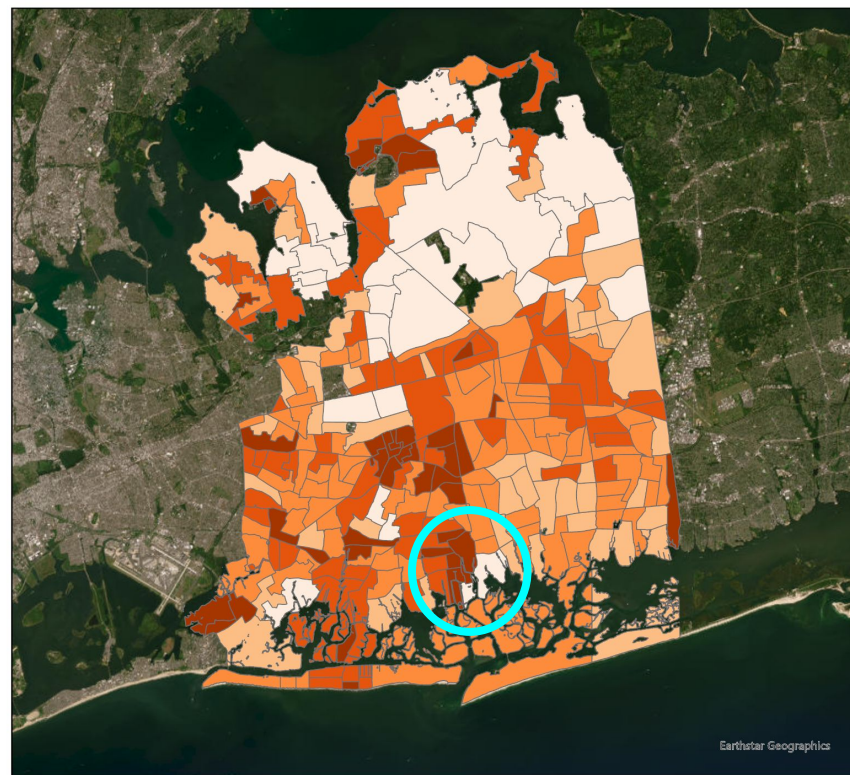


Results



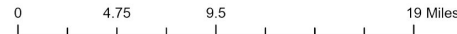
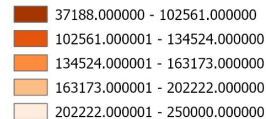
Nassau County Tracts

RoadDensity

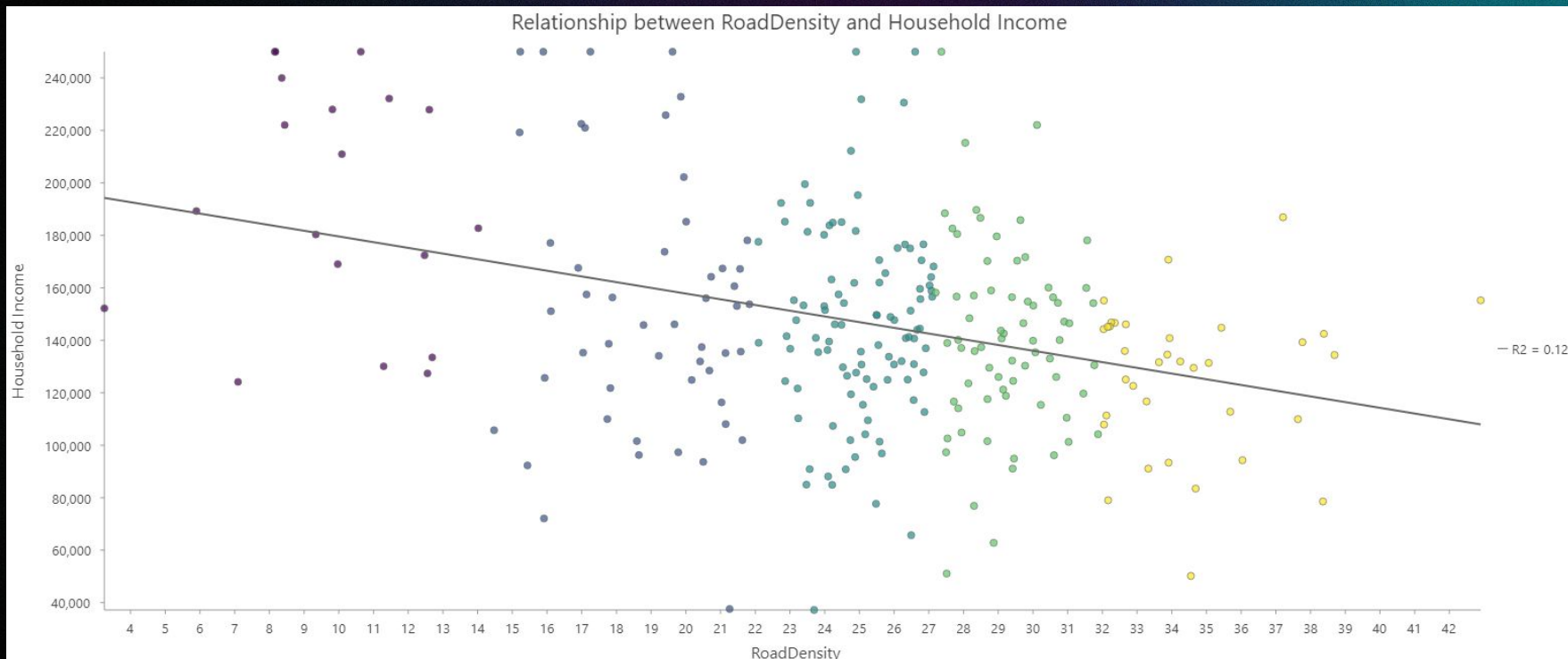


Nassau County Tracts

Household Income



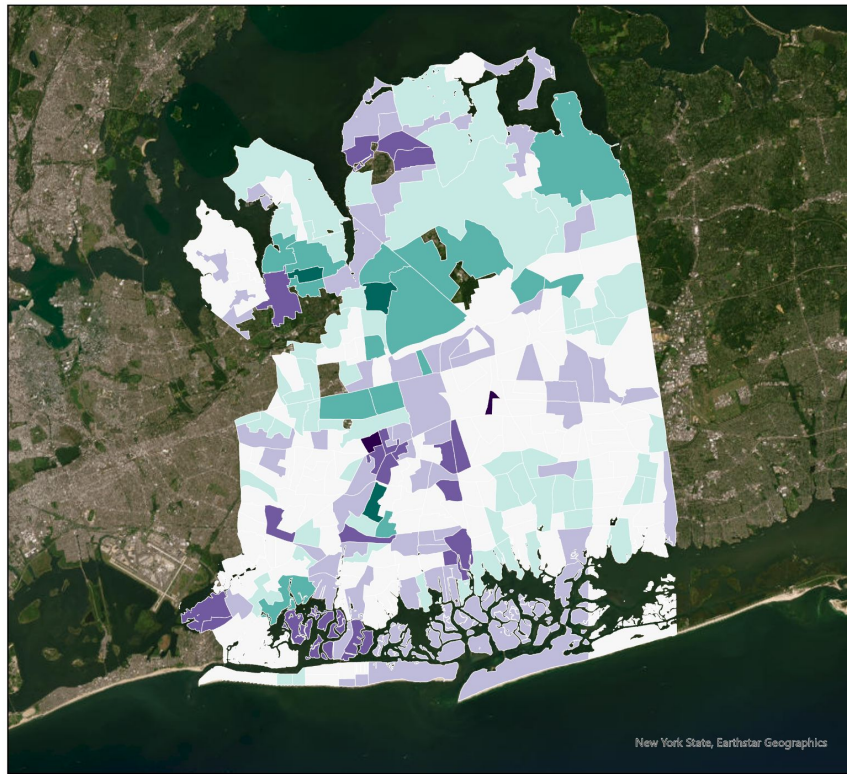
Results



$R^2 = 0.12$ - 12 percent of variance in income is attributable to road density

P-value = 0 - There is a virtually 0% chance of such a result occurring randomly

For every extra mile of roadways per square mile, the estimated household income decreased by \$2700.



Income Vs Density

STD_RESID

< -2.5 Std. Dev.

-2.5 - -1.5 Std. Dev.

-1.5 - -0.5 Std. Dev.

-0.5 - 0.5 Std. Dev.

0.5 - 1.5 Std. Dev.

1.5 - 2.5 Std. Dev.

> 2.5 Std. Dev.



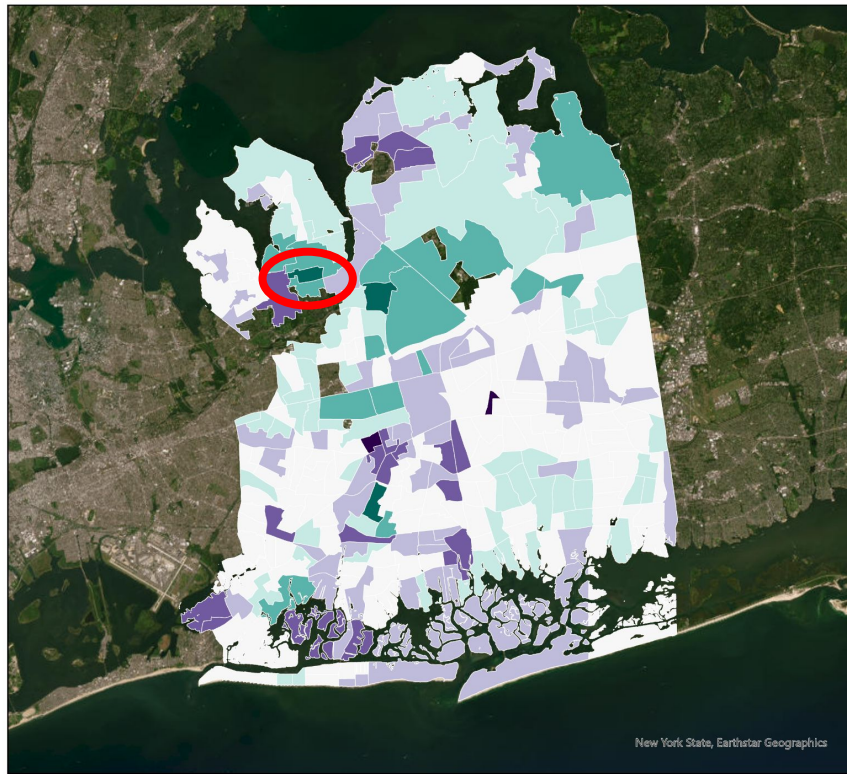
0 4.75 9.5 19 Miles

We can conclude that increased road density is correlated with decreased household income. This does not imply that one causes the other.

Several factors may lead to this result :

Neighborhoods with bigger properties will have less roads and higher household incomes.

Neighborhoods with public parks and golf courses will have areas without roads - and these services will increase property values.



Income Vs Density

STD_RESID

< -2.5 Std. Dev.

-2.5 - -1.5 Std. Dev.

-1.5 - -0.5 Std. Dev.

-0.5 - 0.5 Std. Dev.

0.5 - 1.5 Std. Dev.

1.5 - 2.5 Std. Dev.

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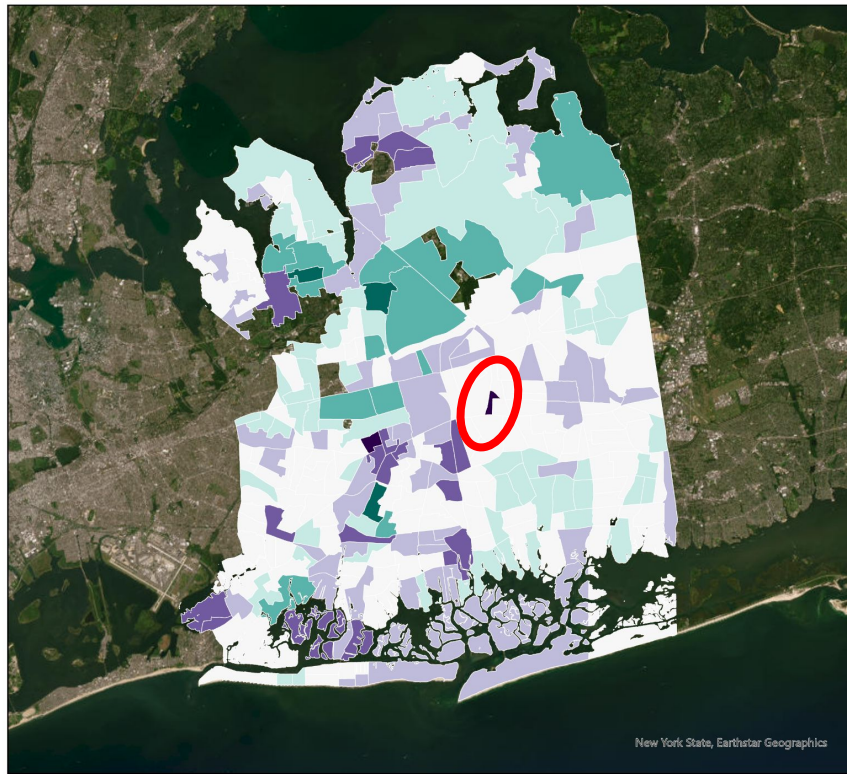


0 4.75 9.5 19 Miles

Policy recommendations:

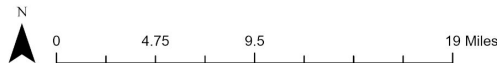
Neighborhoods shown in dark green have much higher road densities than would be expected from a neighborhood with their income.

Munsey Park, highlighted, has a high road density but is one of the wealthiest areas of the country. (13th in 2009). It may be useful for high density suburban neighborhoods to use it as a model for policy.



Income Vs Density

STD_RESID	-1.5 - -0.5 Std. Dev.
	-0.5 - 0.5 Std. Dev.
	0.5 - 1.5 Std. Dev.
	1.5 - 2.5 Std. Dev.
	> 2.5 Std. Dev.
	< -2.5 Std. Dev.
	-2.5 - -1.5 Std. Dev.



Sources of Error and Future research considerations:

Some sources of error include inaccurate road data, missing household income data in certain districts, and inaccurate household income data. Some districts may also not be ideal for a study on suburbs, and may influence the data.

It is also worth noting that some of this correlation may be the cause of redlining and racial segregation, which is a common practice on Long Island.

Future studies may want to assess the existence of this correlation in other suburbs, or in both urban and rural environments.

Thank You.

Slideshow Bibliography

Image sources -

Hempstead village from the air -

By Pi.1415926535 - Own work, CC BY-SA 3.0, <https://commons.wikimedia.org/w/index.php?curid=81285328>

Hempstead House in Sands Point, on Long Island, New York - public domain

All other pictures sourced from own GIS study

Data sources -

<https://data.gis.ny.gov/maps/7d5dff6bcf664e6f83abee9968fc7916/about>

<https://catalog.data.gov/dataset/tiger-line-shapefile-2018-state-new-york-current-census-tract-state-based>

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